Engineering with precision. Innovating with purpose.

RUZZEL A. DULAY, RCE

Civil Engineer | Structural Design · Cost Engineering · Automation

Registered Civil Engineer with experience in structural modeling and analysis support, construction cost management and estimation. I also develop engineering software and automation workflows, with verification and review appropriate to each project.



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Online portfolio · demos, current project details and six targeted resumes
LinkedIn

Engineering background

BS Civil Engineering, TIP–Manila, 2024. PRC licensed November 2024. DOST-SEI Scholar, 2020–2024.

Cost Control Engineer · Millennium Erectors Corporation · April 2026 – Present

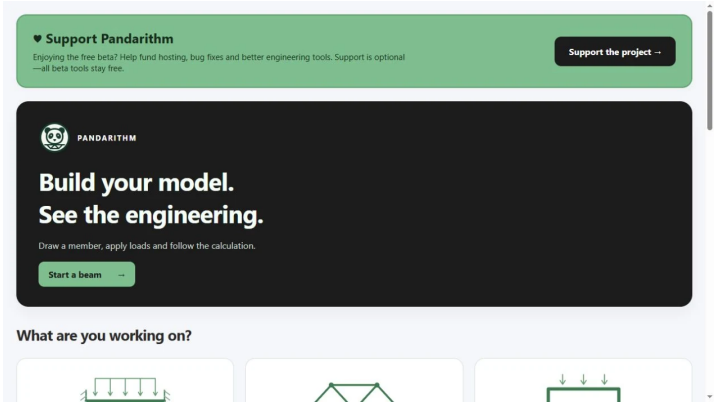
Cost Estimator · International Elevator & Equipment, Inc. · May 2025 – April 2026

Junior Civil Engineer · Alliance Textile Mills Inc. · January – April 2025

This printable selection reflects the current public portfolio. Independent software remains under development; portfolio descriptions do not establish certification, independent design approval or production readiness.

Structural engineering

Pandarithm



Public beta dashboard, captured 21 October 2026.

Objective: Make structural analysis calculations organized, reviewable and repeatable.

Role: Project Creator, Engineering Logic Developer and Validation Lead.

Tools / methods: Python, Flask, NumPy, structural formulations and browser diagrams.

Deliverables: Beta analysis workspaces, diagrams and calculation-report workflows.

Verification: Scoped beam evidence and a live sample checked; module verification is ongoing.

Project details and available demonstrations on the live portfolio
Pandarithm public beta

Thesis, Nonlinear Seismic Analysis: Three Resilience Techniques

Objective: Compare seismic-response strategies for a five-story school.

Role: Thesis researcher and structural-analysis contributor.

Tools / methods: ETABS, SAP2000 and nonlinear time-history comparisons.

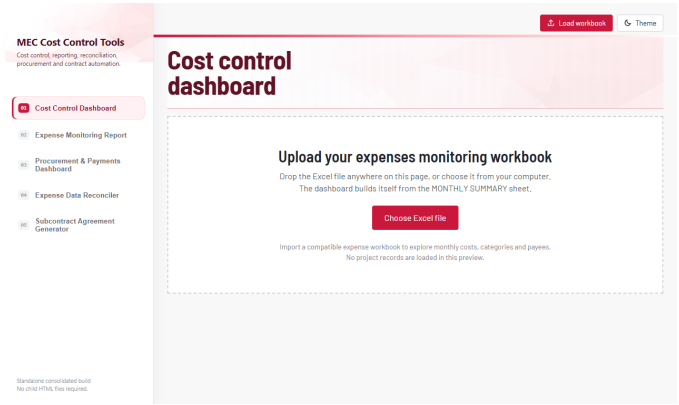
Deliverables: Models and comparative seismic-response findings for the thesis.

Verification: Academic research; not a building safety certification.

Project details and available demonstrations on the live portfolio

Cost engineering & estimation

MEC Cost Control Suite



Actual MEC suite interface. No workbook or project records are loaded in this preview.

Objective: Bring recurring cost reporting and commercial records into a repeatable workflow.

Role: Cost-control workflow and automation development.

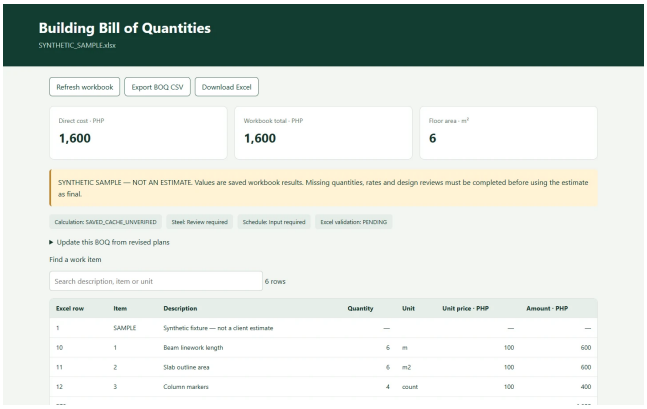
Tools / methods: HTML/JavaScript, Excel workflows, reporting and reconciliation.

Deliverables: Browser suite incorporating dashboard, monitoring, extraction and ledger history.

Verification: Empty interface checked; populated employer records are withheld and outputs require review.

Project details and available demonstrations on the live portfolio

CAD-to-BOQ Estimation Automation



Actual BOQ viewer with isolated synthetic workbook; demonstration quantities and prices, not an estimate.

Objective: Connect reviewed drawing quantities with organized cost-estimation inputs.

Role: Engineering estimation and automation development.

Tools / methods: Python, ezdxf, explicit unit/entity mappings and Excel.

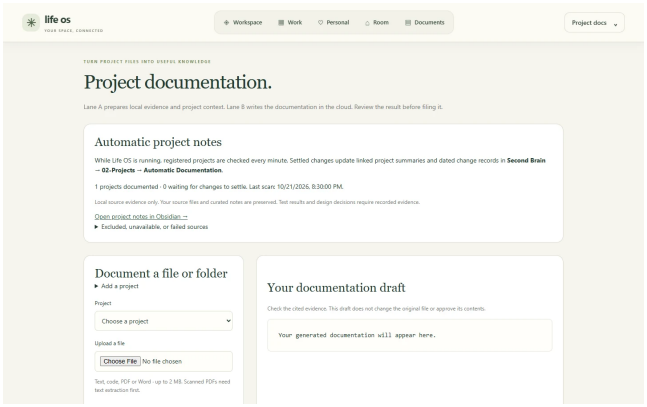
Deliverables: Traceable sample quantity extraction and Excel-connected BOQ workflow.

Verification: Isolated DXF sample checked; complete estimating and native Excel validation remain ongoing.

Project details and available demonstrations on the live portfolio

Engineering automation

Life OS



Actual Life OS documentation interface with isolated sample API data; private vault disconnected.

Objective: Keep project context, documentation and knowledge connected.

Role: Creator of local knowledge and project-documentation workflows.

Tools / methods: Flask, SQLite, Markdown and Obsidian-compatible links.

Deliverables: Local workspace, observed project notes and reviewed documentation workflows.

Verification: Observer tests passed in isolation; private production services were not connected.

Project details and available demonstrations on the live portfolio

Project Lift, Elevator Proposal Automation

Objective: Reduce repeated entry across elevator proposals, specifications and pricing.

Role: Excel/VBA proposal automation developer.

Tools / methods: Excel, VBA, synchronized input and event handling.

Deliverables: Linked proposal, specification and pricing workflow.

Verification: Earlier workplace tool; outputs and assumptions require project review.

Project details and available demonstrations on the live portfolio

IEEI AUTOCALC: Automated Costing & Estimation Tool

Objective: Translate project parameters into repeatable quantity and budget outputs.

Role: Estimation workflow and VBA automation developer.

Tools / methods: Excel, VBA, variable project inputs and linked costing sheets.

Deliverables: Itemized costing and preliminary estimate outputs.

Verification: Earlier workplace tool; not an independently certified estimating system.

Project details and available demonstrations on the live portfolio

Further work & evidence

Project 448, RC Structural Analysis

ETABS modeling and AutoCAD documentation for a reinforced-concrete building.

Tools: ETABS, AutoCAD, RC frame modeling and load cases.
Evidence: Earlier project; independent calculation validation is not established by this portfolio.

Project Steel, Industrial Warehouse Structure

Structural modeling and analysis of an industrial steel warehouse with gravity and lateral load cases.

Tools: ETABS, steel framing, roof trusses, wind and seismic load cases.
Evidence: Earlier project; source scope is described without a new certification claim.

Project RLS, Multi-Story Residential Building

ETABS modeling and structural drawing development for a multi-story residential building.

Tools: ETABS, AutoCAD and reinforced-concrete framing.
Evidence: Earlier project; outputs are shown within the documented scope.

Project QTO, Digital Quantity Take-offs

PlanSwift material takeoffs to support estimation and quantity verification.

Tools: PlanSwift, drawing interpretation and quantity checks.
Evidence: Academic/professional training work; measurement requires reviewed drawings.

Project BOQ, Estimation Workbook Audit & Repair

Schema-based BOQ workbooks and estimation workflows for repeatable civil project planning.

Tools: Excel, workbook schemas and quantity/cost consistency review.
Evidence: Reference build; not a final approved project estimate.

Further work & evidence

Gamified Reviewing Tool

An updated gamified study and review tool, evolved from the earlier review engine.

Tools: Python and HTML/JavaScript, as documented for the earlier implementation.
Evidence: Current revision is under development; no current-version validation claimed.

Coordinated Drafting Workflow

An active coordination workflow bringing drawing sets, structural models, estimates and construction schedules into one reviewable project package.

Tools: Confidential project methods and files are withheld.
Evidence: Confidential; no internal records included.

ETABS Model Audit & Load Automation

An audit-first ETABS workflow for model inventories, explicit group-based loads and deterministic load combinations.

Tools: ETABS-oriented audit logic and local automation scaffold.
Evidence: Local scaffold; production deployment and complete validation not verified.

Calculation Batch Runner & Evidence Reports

Confidential engineering automation project. Implementation details, inputs and reports are withheld.

Tools: Implementation details are withheld.
Evidence: Confidential; no internal records included.
Visit the online portfolio for full case studies and captioned demos